

Machine learning methods for continuous glucose monitoring data and their potential in cardiovascular risk assessment

PhD dissertation

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Disclosures and Notes

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Notes on website version

This dissertation is available for viewing online at HeleneThomsen.github.io/dissertation/ or by scanning the QR code in the lower right-hand corner of the title page.

Acknowledgements

Preface

Outline of the dissertation

Papers in the dissertation

Study I

Thomsen, H. B., Lebiecka-Johansen, B., Nørgaard, O., Andersen, T. H., Andersen, S. T., Fagherazzi, G., Hulman, A., & Isaksen, A. A. (2026). Continuous Glucose Monitoring–Derived Metrics and Cardiovascular Risk Among People With Diabetes: Systematic Scoping Review. *JMIR Diabetes*, 11(1), e89374. <https://doi.org/10.2196/89374>

Study II

Thomsen, H. B., Andersen, S. T., Fagherazzi, G., Isaksen, A. A. & Hulman, A. (2026). Foundation model–derived representations of continuous glucose monitoring data: relation to glycemic patterns and cardiovascular risk. (Under review at *JMIR AI*, (7.30.2026), pending for preprint at *JMIR preprints*)

Study III

Thomsen, H. B., Li, L. Y., Isaksen, A. A., Lebiecka-Johansen, B., Bour, C., Fagherazzi, G., Doorn, W. P. T. M. van, Varga, T. V., & Hulman, A. (2025). Racial disparities in continuous glucose monitoring-based 60-min glucose predictions among people with type 1 diabetes. *PLOS Digital Health*, 4(6), e0000918. <https://doi.org/10.1371/journal.pdig.0000918>

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Abbreviations

AGP: ambulatory glucose profile

AID: Automated insulin delivery

CGM: continuous glucose monitoring

CV: coefficient of variation

CVD: cardiovascular disease

eGFR: Estimated Glomerular Filtration Rate

GMI: glucose management indicator

HbA1c: hemoglobin A1c

MBG: mean glucose

NT-proBNP: N-terminal Pro-B-type Natriuretic Peptide

SCORE2-Diabetes: Systematic Coronary Risk Evaluation 2-Diabetes

SMBG: self-monitoring blood glucose

T1D: type 1 diabetes

T2D: type 2 diabetes

TAR: time above range

TBR: time below range

TIR: time in range

Introduction 1

1.1 Diabetes

Diabetes is a metabolic disease characterized by high blood glucose levels known as hyperglycemia.^{1,2} The disease is one of the fastest growing global health challenges.³ In 2021 it was estimated that 537 million people lived with diabetes, and it is expected to reach 643 million by 2030.³ The most common forms of chronic diabetes are type 1 diabetes (T1D) and type 2 diabetes (T2D).¹ T1D is an autoimmune disease characterized by the destruction of the pancreatic islet β -cells, leading to an absolute insulin deficiency.⁴ Approximately 85%-95% of all people with diabetes in developing countries have T2D.¹ It is the most common form of diabetes and in contrast to T1D it generally begins with insulin resistance in which the body's cells become less responsive to insulin.^{1,5} Initially, the pancreas compensates by increasing insulin secretion, but over time progressive β -cell dysfunction results in inadequate insulin production to meet the body's demands.^{1,2,5} T2D is the most common form of diabetes. Despite their distinct underlying pathophysiology, both T1D and T2D are characterized by chronic hyperglycemia.

Diabetes can be diagnosed based on a bloodtest to check hemoglobin A1c (HbA1c) levels or through plasma glucose criteria (Table 1.1).^{4,6}

Although both plasma glucose tests and HbA1c tests are appropriate for diabetes screening, the HbA1c test has several advantages.⁴ Some of the benefits are that it requires no fasting and it is less sensitive to stress and changes in nutrition.⁴ HbA1c is an indirect measure of blood glucose levels because glucose binds hemoglobin over ~ 120 days lifespan of erythrocytes (red blood cells).⁴ However HbA1c testing is not always accessible, affordable or suitable (e.g., for people with anemia or under HIV-treatment).⁴ With the guidelines recommendation of using HbA1c, a debate has sparked on the controversy of using a set thresholds as it is well known that racial differences in HbA1c exists.⁷ Black individuals have higher HbA1c levels than Hispanic and non-Hispanic White people with diabetes,^{4,8} however studies show that the association between HbA1c and risk of complications appear to be similar between ethnic race groups.⁴

1.2 Blood Glucose Monitoring

HbA1c values are a very important part of diabetes management. The treatment goal is for HbA1c to be below 53 mmol/mol for non-pregnant adults without any hyperglycemia or hypoglycemia (low blood glucose levels). However HbA1c goals should be individualized to fit each individual's specific situation. HbA1c can vary from e.g., <48 mmol/mol to none.^{9,10} There are two tools available for self-assessment of glycemia, the self-monitoring blood glucose (SMBG) device measuring capillary blood with finger-sticks, and the continuous glucose monitoring (CGM) measuring interstitial fluid continuously.^{10,11} Both monitoring devices have shown to improve glycemic control in people with diabetes, however the CGM is superior when it comes to immediate information and alerts on asymptomatic hypo- and hyperglycemic events and

Table 1.1: Diagnostic tests for diabetes

Test	Criteria	Description
Fasting plasma glucose	≥ 7.0 [mmol/L]	Fasting is defined as no calorie intake for at least eight hours.
2-hour plasma glucose	≥ 11.1 [mmol/L]	Glucose is measured 2 hours after oral ingestion of 75g glucose dissolved in water (OGTT)
Random plasma glucose	≥ 11.1 [mmol/L]	Individuals with classic symptoms of hyperglycemia (e.g., dehydration and unintentional weight loss) or hyperglycemic crisis (e.g., diabetic ketoacidosis) together with a plasma test taken any time of day.
Glycated hemoglobin (HbA1c)	≥ 48 [mmol/mol]	

provide a comprehensive glucose profile including both pre- and postprandial blood glucose measurements.^{10,12,13}

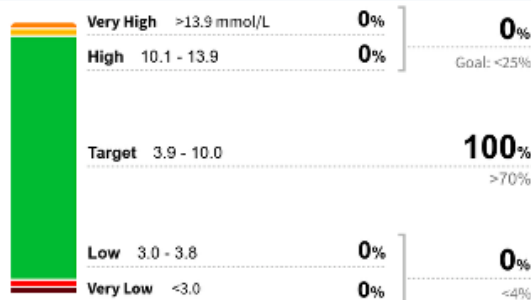
The CGM devices measures the glucose in the interstitial fluid, which is the fluid around and between cells. It corresponds with plasma glucose although a lag can appear when glucose levels rise or fall rapidly.¹⁴ Use of CGM devices has shown to improve glycemic control in both T1D and T2D with improvements in HbA1c levels and more time spent in target range (3.9-10 mmol/L) regardless of age, sex, education and income.^{9,14} CGM devices focuses on monitoring and it is possible to get an ambulatory glucose profile (AGP) report from you device (Figure 1.1). The AGP report shows summary statistics over all your data (usually a two weeks period) including CGM-derived metrics as shown in (Table 1.2).¹² Besides monitoring, data from CGM devices are now used in combination with insulin delivery systems to automate the process of administering insulin.¹⁴ Automated insulin delivery (AID) systems typically use data from CGM devices to determine or adjust insulin delivery. The simplest AIDs are threshold-based low-glucose suspend systems in which insulin delivery is automatically suspended when blood glucose levels drop below a predefined threshold.^{race7?} More advanced systems utilize additional information such as carbohydrate intake and physical activity to optimize insulin delivery.^{race8?}

Glucose Pattern Insights

Selected Dates: 11 Apr 2024 - 24 Apr 2024 (14 Days)

LibreView

Time in Ranges



Glucose Statistics

Average Glucose

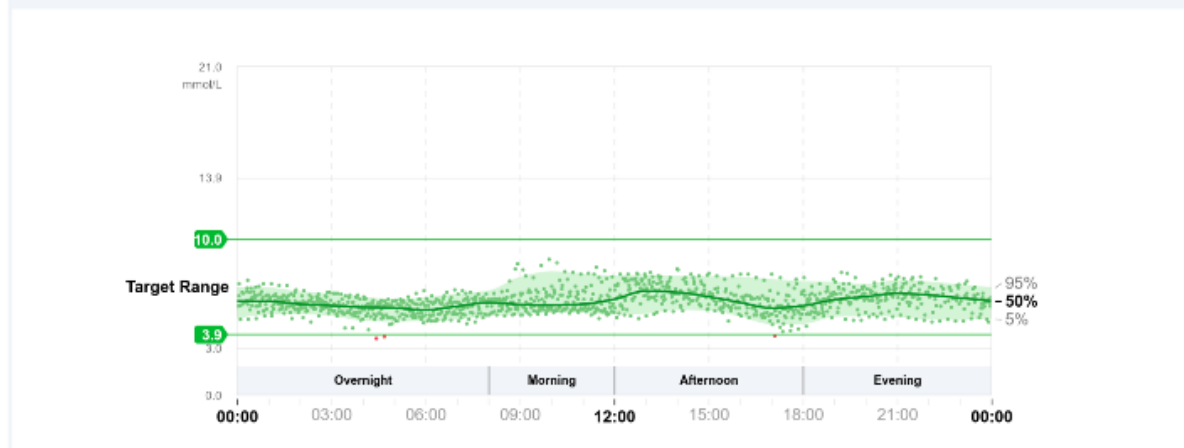
6.0 mmol/L Goal: ≤8.6 mmol/L

Glucose Management Indicator (GMI)

Approximate A1C level based on average CGM glucose level.

5.9% Goal: ≤7.0% | 41 mmol/mol Goal: ≤53 mmol/mol

Glucose Patterns (14 Days)



DAILY GLUCOSE PROFILES

Each daily profile represents a midnight to midnight period with the date displayed in the upper left corner.

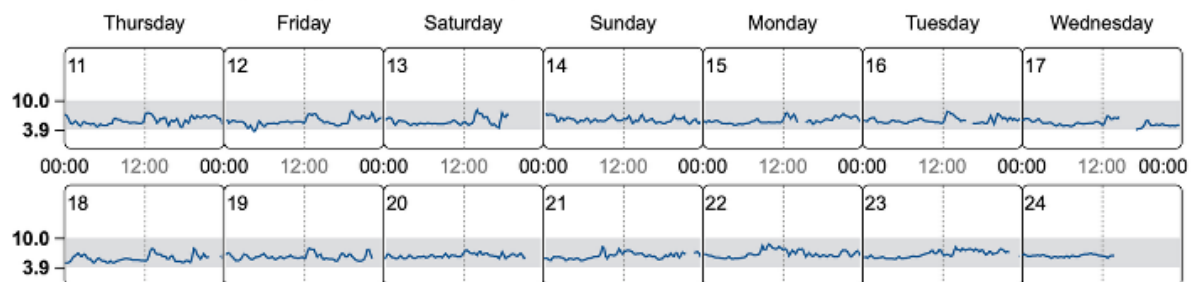


Figure 1.1: Ambulatory glucose profile report from female without diabetes. Data points are from the author herself using an intermittently-scanned (Manuel data upload) FreeStyle LibreLink sensor.

1.3 Glycemic Variability

The challenge in obtaining good glycemic control is to keep blood glucose levels low enough to prevent comorbidities while avoiding the acute symptoms and, for insulin treated people with diabetes, potential mortality of iatrogenic (treatment-induced) hypoglycemia.¹⁵ The variability in the blood glucose fluctuations needs to be reduced in order to lower the average blood glucose levels while still avoiding hypoglycemia.¹⁵ Advances in CGM technology have enabled the quantification of glucose variability, a measure of the extent of fluctuations in blood glucose levels over time,

Table 1.2: CGM-derived metrics presented in the ambulatory glucose profile report with target values.

Abbreviation	CGM-derived metric	Range	Recommended target
GMI	Glucose Management Indicator		No specific target
TIR	Time in Range	3.9-10.0 mmol/L (70-180 mg/dL)	> 70%
TBR ₅₄	Time Below Range	< 3.0 mmol/L (< 54 mg/dL)	< 1%
TBR ₇₀	Time Below Range	< 3.9 mmol/L (< 70 mg/dL)	< 4%*
TAR ₁₈₀	Time Above Range	> 10.0 mmol/L (> 180 mg/dL)	< 25%**
TAR ₂₅₀	Time Above Range	> 13.9 mmol/L (> 250 mg/dL)	< 5%
CV	Coefficient of Variation		≤ 36%
	Number of days sensor was worn		≥ 14 days for pattern management
	Active sensor time		70% of 14 days

* This goal include both TBR₅₄ and TBR₇₀.

** This goal include both TAR₁₈₀ and TAR₂₅₀

including both peaks and nadirs.¹⁶ Describing the variability is clinically relevant because two individuals with similar HbA1c levels may exhibit different glycemic variability profiles.¹⁵ Numerous metrics have been developed to characterize different aspects of glucose variability, but no single measure captures all of its dimensions. Consequently, no universal consensus has been reached, although the coefficient of variation (CV) and SD have been incorporated into AGP reports and the American Diabetes Association guidelines (Table 1.2).^{15,16}

1.4 Cardiovascular Diseases and Predicting Risk

Although advances in diabetes management have reduced the incidence of several diabetes-related complications, cardiovascular disease remains a major cause of morbidity and mortality in people with diabetes.¹⁷⁻¹⁹ Chronic hyperglycemia can lead to long-term complications, although the progression and manifestation of these complications vary between individuals.⁴ Identifying individuals at increased risk of atherosclerotic cardiovascular disease or heart failure before clinical disease develops is therefore an important component of diabetes care.¹⁸

Risk prediction models estimating an individual's future risk of developing cardiovascular disease (CVD) have been developed and are widely used.¹⁹⁻²² By estimating the probability of a CVD event within a specified time horizon, these models can increase awareness of cardiovascular risk, encourage adherence to recommended lifestyle changes and medical treatment, and help identify individuals who are most likely to benefit from preventive interventions.²³ Among these, the Systematic Coronary Risk Evaluation 2-Diabetes (SCORE2-Diabetes) model estimates the 10-year risk of CVD in individuals with T2D. It was developed and calibrated using European cohorts, making it particularly relevant for the Danish cohort included in this dissertation.¹⁹ However, the SCORE2-Diabetes model relies solely on traditional clinical variables, including

age, sex, smoking status, systolic blood pressure, high-density lipoprotein cholesterol, age at diabetes diagnosis, HbA1c, and Estimated Glomerular Filtration Rate (eGFR), and does not incorporate information on glucose dynamics beyond the long-term glycemic exposure reflected by HbA1c.¹⁹ HbA1c have been independently associated with risk of heart failure,²⁴ however it does not capture the short-term glucose fluctuations, which may be associated with the progression of vascular complications in diabetes.²⁵ Associations between CGM-derived metrics and CVD outcomes have been investigated in the literature^{26–28} however evidence regarding the predictive power of CGM-derived metrics in predicting CVD remains limited.²⁹

1.5 Heart Failure

The limited evidence on the predictive value of CGM data is particularly important for heart failure, as the condition is at least twice as prevalent among people with diabetes as in the general population.^{18,30} Heart failure may present as either an acute or a chronic condition.³¹ It is a clinical syndrome characterized by a combination of symptoms, including breathlessness, fatigue, and ankle swelling, together with clinical signs such as elevated jugular venous pressure and pulmonary crackles.³¹ In individuals with suspected chronic heart failure based on risk factors, symptoms, clinical findings, and supporting investigations such as electrocardiography, measurement of N-terminal Pro-B-type Natriuretic Peptide (NT-proBNP) is recommended as part of the diagnostic work-up.³¹ An NT-proBNP concentration of 125 pg/mL indicates the need for further cardiac evaluation, whereas concentrations below this threshold make chronic heart failure unlikely and alternative diagnoses should be considered.³¹ Consequently, NT-proBNP represents a clinically relevant biomarker for identifying individuals at increased risk of chronic heart failure.

1.6 Association vs Prediction

Many studies have

1.7 Machine learning

Relevant? Artificial Intelligence: The Future for Diabetes Care: <https://www.clinicalkey.com/#!/content/playContent/1-s2.0-S0002934320303399?returnurl=https:%2F%2Flinkinghub.elsevier.com/doi/10.1016/j.jdi.2023.100000>
 Mo2023: says something about glucose variability metrics, what they are and what critiques are there to say about them: Glycemic variability: Measurement, target, impact on complications of diabetes and does it really matter? <https://onlinelibrary.wiley.com/doi/10.1111/jdi.14112>

- Data-driven vs hypothesis-driven(knowledge-based) paradigms
- Why machine learning in diabetes and cardiovascular risk prediction?
- Why are ML fantastic for CGM data?
- Types of machine learning methods
 - Logistic regression
 - LSTM and CNNs
 - Foundation models and their advantage
 - The CGMformer is bla bla bla³²
- Challenges and pitfalls of machine learning in healthcare
 - Overfitting and generalizability
 - Interpretability and explainability

1.8 Fairness and bias in machine learning

- Define race and ethnicity
- Algorithmic bias and group fairness
- Fairness metrics and mitigation strategies
- Trends in the quality of care and racial disparities in Medicare managed care:
<https://pubmed.ncbi.nlm.nih.gov/16107622/>

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Study I: Search strings A

Table A.1: Search in Ovid MEDLINE (Search date: 22 January 2024)

#	Searches	Results
1	Blood Glucose	10241
2	Self-Monitoring/ Continuous Glucose Monitoring/	34
3	((blood-glucose or bloodglucose or glucose) adj2 (monitor\$ or selfmonitor)) <i>or</i> cgmorb ^{gmoriscgmorr} tcgmor(<i>timeadj3range</i>)). <i>ti, ab, kf, kw</i> or bloodsugar\$ or glucose\$ or glyc?emic\$ or sugar). <i>ti, ab, kf, kw.</i>)adj5(<i>flashorfluctuat</i> or intermittent\$ or scan\$ or variabilit). <i>ti, ab, kf, kw.</i> 8669 5 <i>or/1—</i> 4 38366 6 <i>expCardiovascularDiseases/</i> 2760764 7 (cardiovasc or cvd or cardiometabol\$ or cardio-metabol\$ or heart\$ or macrovasc\$ or vasc). <i>ti, ab, kw, kf.</i> 2182473 8 <i>or/6—</i> 7 3971342 9 <i>expDiabetesMellitus/</i> 518864 10 (diabet or dm1 or dm2 or dmi or dmii or iddm or niddm or t1d or t2d). <i>ti,ab,kw,kf.</i> or/9-10	815092
11	or/9-10	875527
12	and/5,8,11	2416
13	exp Animals/ not Humans/	5189493
14	12 not 13 Updated search: 11 March 2025	2300
15	limit 14 to (da="20231201-20261231" or dt="20231201-20261231" or ez="20231201-20261231" or ed="20231201-20261231")	310
16	or/14-15	2610

Table A.2: Search in Embase (Ovid) (Search date: 22 January 2024)

#	Searches	Results
1	blood glucose monitoring/	37574
2	exp continuous glucose monitoring system/	5648
3	((blood-glucose or bloodglucose or glucose) adj2 (monitor\$ or selfmonitor)) <i>or</i> <i>cgmorb</i> <i>gmoris</i> <i>cgmorr</i> <i>tcgmor</i> (<i>timeadj3range</i>)). <i>ti, ab, kf, kw.</i> 4850 or bloodsugar\$ or glucose\$ or glyc?emic\$ or sugar). <i>ti, ab, kf, kw.</i>) <i>adj5</i> (<i>flashorfluctuat</i> or intermittent\$ or scan\$ or variabilit). <i>ti, ab, kf, kw.</i> 16430 5 <i>or</i> /1–4 81510 6 <i>expcardiovascular</i> <i>disease</i> / 5122163 7 (cardiovasc or cvd or cardiometabol\$ or cardio-metabol\$ or heart\$ or macrovasc\$ or vasc). <i>ti, ab, kw, kf.</i> 3044925 8 <i>or</i> /6–7 6303633 9 <i>expdiabetesmellitus</i> / 1239184 10 (diabet or dm1 or dm2 or dmi or dmii or iddm or niddm or t1d or t2d). <i>ti,ab,kw,kf.</i>	1237019
11	or/9-10	1501812
12	and/5,8,11	11217
13	(rat or rats or mouse or mice or swine or porcine or murine or sheep or lambs or pigs or piglets or rabbit or rabbits or cat or cats or dog or dogs or cattle or bovine or monkey or monkeys or trout or marmoset\$1). <i>ti.</i> and animal experiment/	1237966
14	animal experiment/ not (human experiment/ or human/)	2601109
15	or/13-14	2672240
16	12 not 15	10861
17	limit 16 to exclude medline journals Updated search: 11 March 2025	2176
18	limit 17 to (dc=“20231201-20261231” or dd=“20231201-20261231” or rd=“20231201-20261231”)	467
19	or/17-18	2643

Study I: Included and excluded reports **B**

C Study I: Geographical location

Asia (n=27) • China (n=19) • Japan (n=8) Europe (n=22) • Italy (n=7) • 3 Belgium (n=3) • 2 Ukraine (n=2) • 2 Finland (n=2) • 2 Spain (n=2) • 1 Greece (n=1) • 1 Malta (n=1) • 1 Sweden (n=1) • 1 The Netherlands (n=1) • 1 Portugal (n=1) • 1 Russia (n=1) North America (n=2) • 2 USA (n=2) Australia / Oceania (n=1) • 1 Australia (n=1)

Borg 2011(ADAG study) had study centers in: • Europe: Netherlands, Denmark, Italy • Africa: 2011 • North America: USA

Borg R, Kuenen J C, Carstensen B, Zheng H, Nathan D M, Heine R J, et al. HbA1(c) and mean blood glucose show stronger associations with cardiovascular disease risk factors than do postprandial glycaemia or glucose variability in persons with diabetes: the A1C-Derived Average Glucose (ADAG) study. 2011; ADAG Study Group, editor. Diabetologia. 54: 69–72. doi:10.1007/s00125-010-1918-2

Study I: Subclinical results

D

- ☐ Add table with subclinical study design overviews
- ☐ Add table overview of subclinical results

E Study II: Prediction stability

Prediction stability plots (?@fig-pred__stability) revealed more stability the fewer predictors.

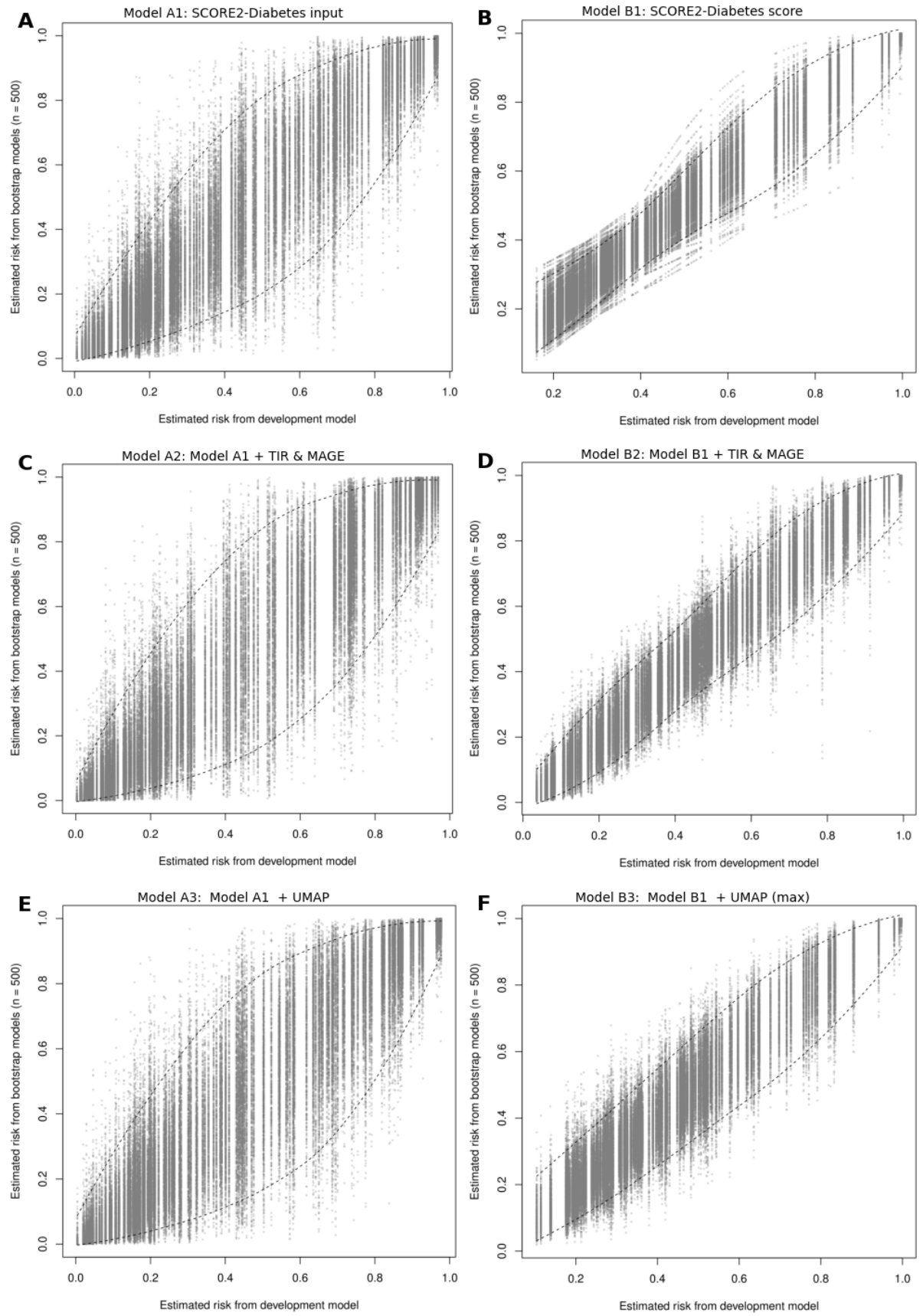


Figure E.1: Prediction stability plots show the predictions (estimated risk) for the full dataset from the development model (equivalent to the apparent model) on the x-axis, and the predictions from all bootstrap models on the y-axis. Tight alignment along the diagonal indicates higher prediction stability.

F Study III: Surveillance Error Grid Results

Table F.1: Results from the Surveillance Error Grid for White participants. Values are reported as percentages.

Group	Model	Zone	0	20	40	60	80	100
White	LOCF	None	69%	69%	69%	69%	69%	69%
		Slight	26%	26%	26%	26%	26%	26%
		Moderate	4%	4%	4%	4%	4%	4%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Individual	None	66%	66%	66%	66%	66%	66%
		Slight	28%	28%	28%	28%	28%	28%
		Moderate	6%	6%	6%	6%	6%	6%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Generalized	None	72%	73%	73%	73%	73%	73%
		Slight	25%	25%	25%	24%	24%	24%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Transfer Learned	None	72%	72%	73%	73%	73%	73%
		Slight	25%	24%	24%	24%	24%	24%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%

□ maybe add this one to appendix, since I think they show the same things

Table F.2: Results from the Surveillance Error Grid for Black participants. Values are reported as percentages.

Group	Model	Zone	0	20	40	60	80	100
Black	LOCF	None	71%	71%	71%	71%	71%	71%
		Slight	24%	24%	24%	24%	24%	24%
		Moderate	4%	4%	4%	4%	4%	4%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Individual	None	66%	66%	66%	66%	66%	66%
		Slight	27%	27%	27%	27%	27%	27%
		Moderate	6%	6%	6%	6%	6%	6%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%

Group	Model	Zone	0	20	40	60	80	100
	Base -Generalized	Extreme	0%	0%	0%	0%	0%	0%
		None	75%	74%	75%	75%	74%	75%
		Slight	23%	23%	22%	22%	22%	22%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
	Transfer Learned	Extreme	0%	0%	0%	0%	0%	0%
		None	75%	74%	75%	75%	74%	75%
		Slight	22%	23%	22%	22%	22%	22%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%

G Full papers

G.1 Paper I

G.2 Paper II

G.3 Paper III

Addenda and errata



This chapter contains an overview of:

- Addenda: contents added since the submitted version of the dissertation.
- Errata: corrections to errors discovered since submission of the dissertation.

H.1 Addenda

H.2 Errata